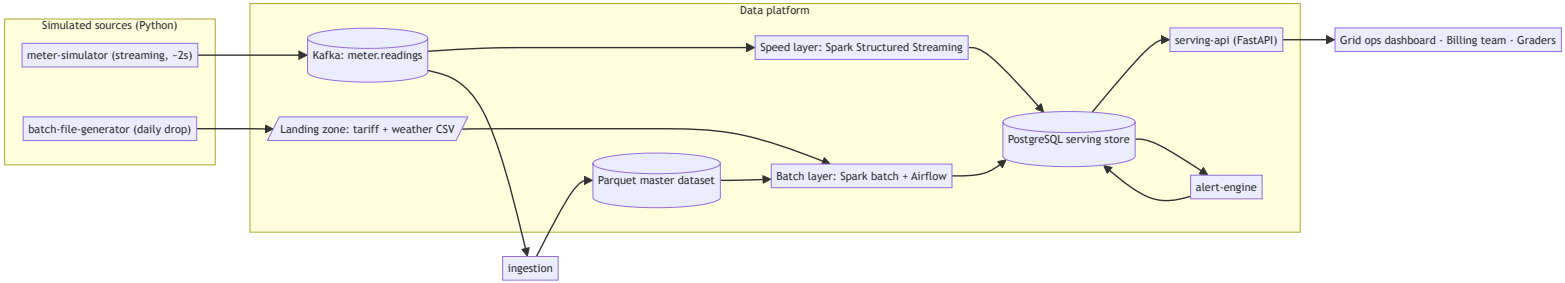
Simulated sources (Python)
meter-simulator (streaming, ~2s)
batch-file-generator (daily drop)
Data platform
Kafka: meter.readings
Landing zone: tariff + weather CSV
Speed layer: Spark Structured Streaming
Batch layer: Spark batch + Airflow
Parquet master dataset
ingestion
PostgreSQL serving store
serving-api (FastAPI)
alert-engine
Grid ops dashboard - Billing team - Graders